

CS:4980

Foundations of Embedded Systems

The Asynchronous Model

Part I

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Asynchronous Model

Recall: In the Synchronous Model, all components execute **in lock-step** in a sequence of (logical) rounds

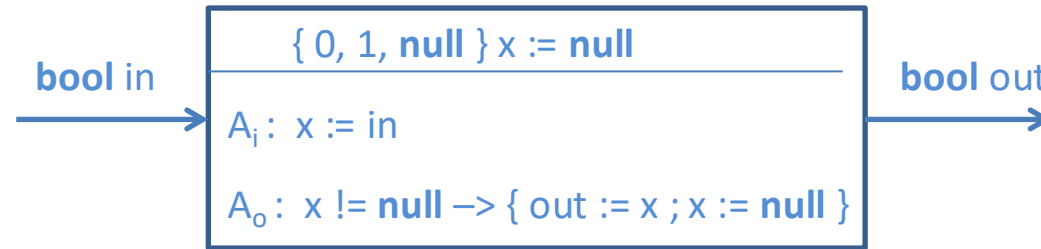
In the **Asynchronous Model** instead the **speeds** at which different components execute are **independent**, or unknown

Examples:

- Processes in a distributed system
- Threads in a typical operating system

Key design challenge: how to achieve coordination

Example: Asynchronous Buffer



Input channel: **in** of type **bool**

Output channel: **out** of type **bool**

State variable: **x**; can be empty (**null**) or hold **0/1** value

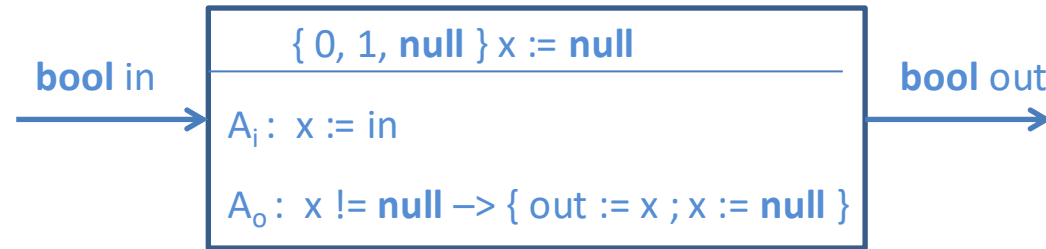
Initialization: **x := null**

Input task: A_i processing input: **x := in**

Output task: A_o producing outputs:

Guard: **x != null** **Update:** **out := x ; x := null**

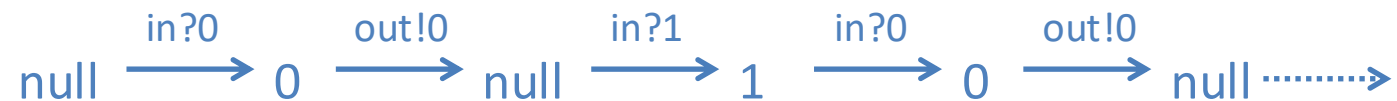
Example: Asynchronous Buffer



Execution Model: only **one task per step** is executed

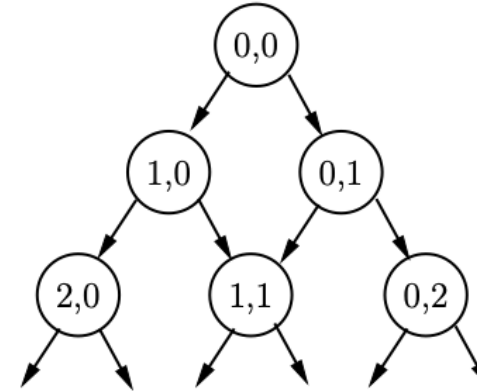
- processing of inputs (by input tasks) is decoupled from production of outputs (by output tasks)
- A task can be executed if it is **enabled**, i.e., its guard holds or is missing
- If multiple tasks are enabled, one of them is executed non-deterministically

Possible execution:



Example: Asynchronous Increments

<code>nat x := 0 ; y := 0</code>
<code>A_x : x := x + 1</code>
<code>A_y : y := y + 1</code>



An *internal task* does not involve input or output channels

- can have guard condition and update code
- is executed in an internal action

In each step, execute, either task A_x or task A_y

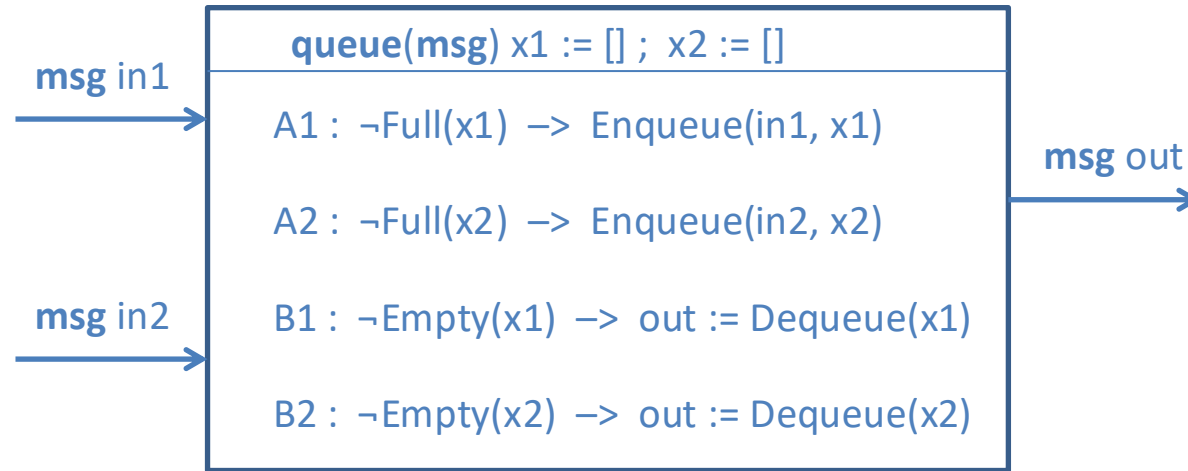
Possible execution:

$(0,0) \rightarrow (1,0) \rightarrow (1,1) \rightarrow (1,2) \rightarrow (1,3) \rightarrow \dots \rightarrow (1,105) \rightarrow (2, 105) \rightarrow \dots$

For every m, n , state $\{ x := m, y := n \}$ is reachable

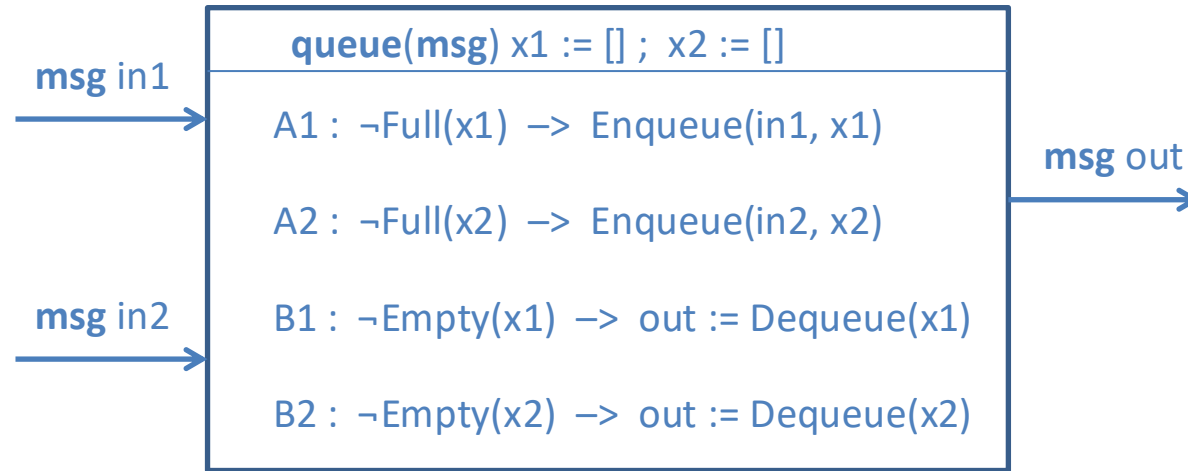
- *Interleaving* model of concurrency, where each task's **update** is **atomic**

Asynchronous Merge



Sequence of messages on output channel is an **arbitrary** merge of sequences of values on the two input channels

Asynchronous Merge

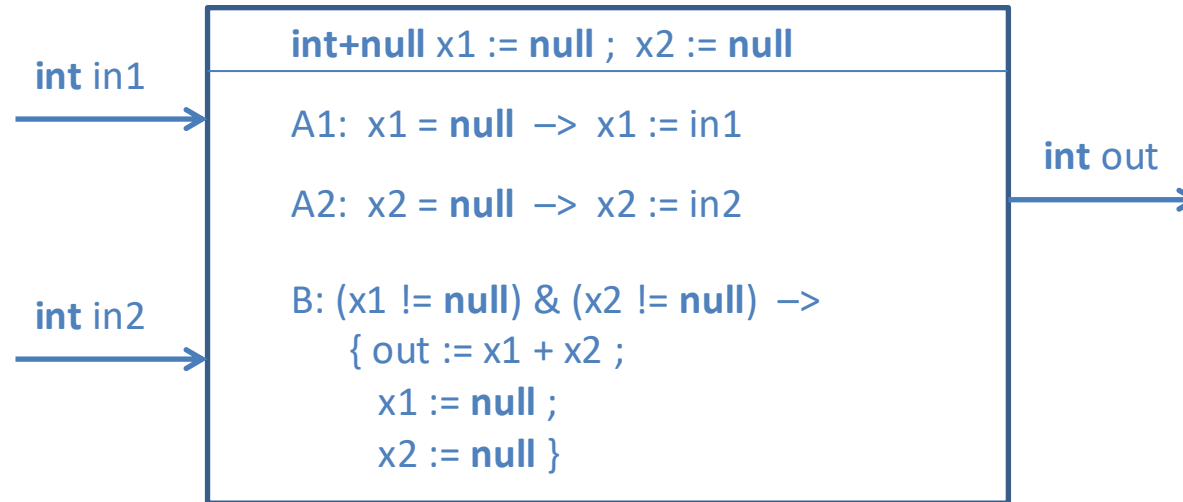


At every step, **exactly one** of the four tasks executes, provided its guard holds

Possible execution:

$([], []) \rightarrow ([5], []) \rightarrow ([5], [0]) \rightarrow ([5], []) \rightarrow ([5,6], []) \rightarrow ([5,6], [3]) \rightarrow ([6], [3]) \rightarrow \dots$
out: // // 0 // // 5

Exercise: What does this component do?



Components are now called *processes*

Asynchronous Process P

Set I of (typed) *input channels*

- Defines the set of inputs of the form $x?v$, where x is an input channel and v a value

Set O of (typed) *output channels*

- Defines the set of outputs of the form $y!v$, where y is an output channel and v a value

Set S of (typed) *state variables*

- Defines the set of states Q_s

An *initialization* $Init$

- Defines the set of initial states

Asynchronous Process P (cont.)

Set of *input tasks*, each associated with an input channel x

- Guard condition over *state variables* S
- Update code from *read-set* $S \cup \{x\}$ to *write-set* S
- Defines a set of *input actions* of the form $s - x?v \rightarrow t$

Set of *output tasks*, each associated with an output channel y

- Guard condition over *state variables* S
- Update code from *read-set* S to *write-set* $S \cup \{y\}$
- Defines a set of *output actions* of the form $s - y!v \rightarrow t$

Set of *internal tasks*

- Guard condition over *state variables* S
- Update code from *read-set* S to *write-set* S
- Defines a set of *internal actions* of the form $s - \varepsilon \rightarrow t$

Asynchronous Gates



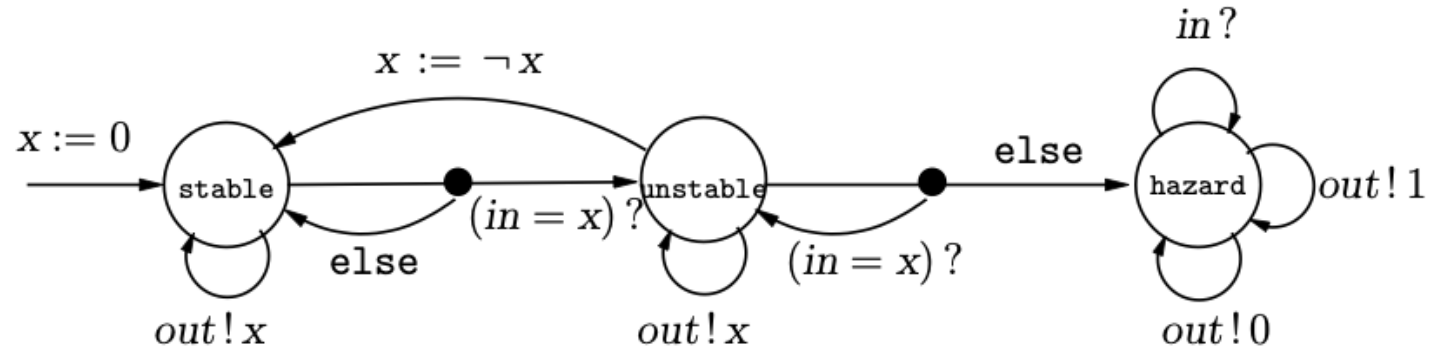
Why design **asynchronous circuits**?

- Input can be changed even before the effect propagates through the entire circuit
- Can be faster than synchronous circuits, but design is more complex

Example: modeling a **NOT** gate

- When input changes, gate enters *unstable* state until it gets a chance to update its output value
- Later input changes in unstable state lead to a *hazard* state with unpredictable behavior

Asynchronous NOT Gate as an ESM



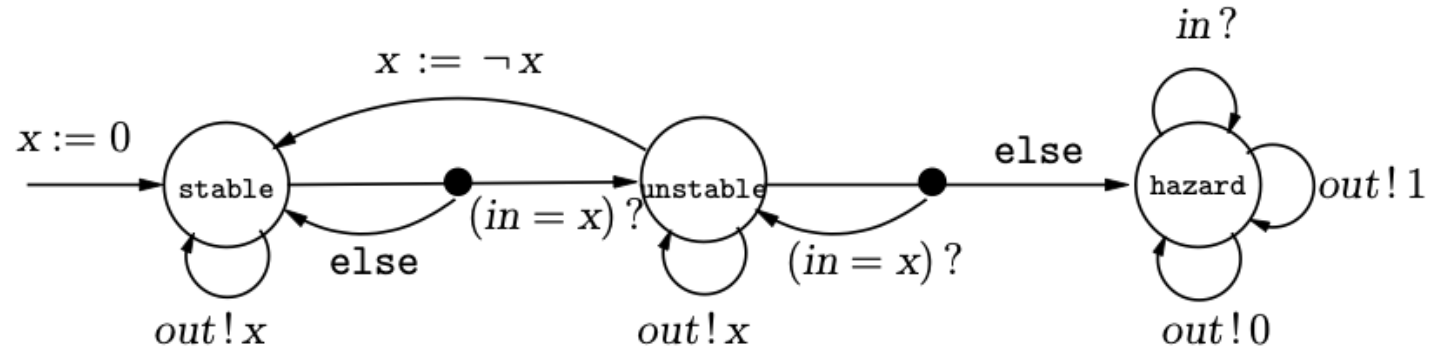
Possible execution:

$(stable, 0) \xrightarrow{out!0} (stable, 0) \xrightarrow{in?0} (unstable, 0) \xrightarrow{\varepsilon} (stable, 1) \xrightarrow{out!1} (stable, 1) \xrightarrow{in?1} (unstable, 1) \xrightarrow{out!1} (unstable, 1) \xrightarrow{in?0} (hazard, 1) \xrightarrow{out!0} (hazard, 1) \xrightarrow{out!1} (hazard, 1) \dots$

How to ensure that the gate does not enter hazard state?

- Environment should wait to see a change in value of output before toggling input again

Executing an ESM

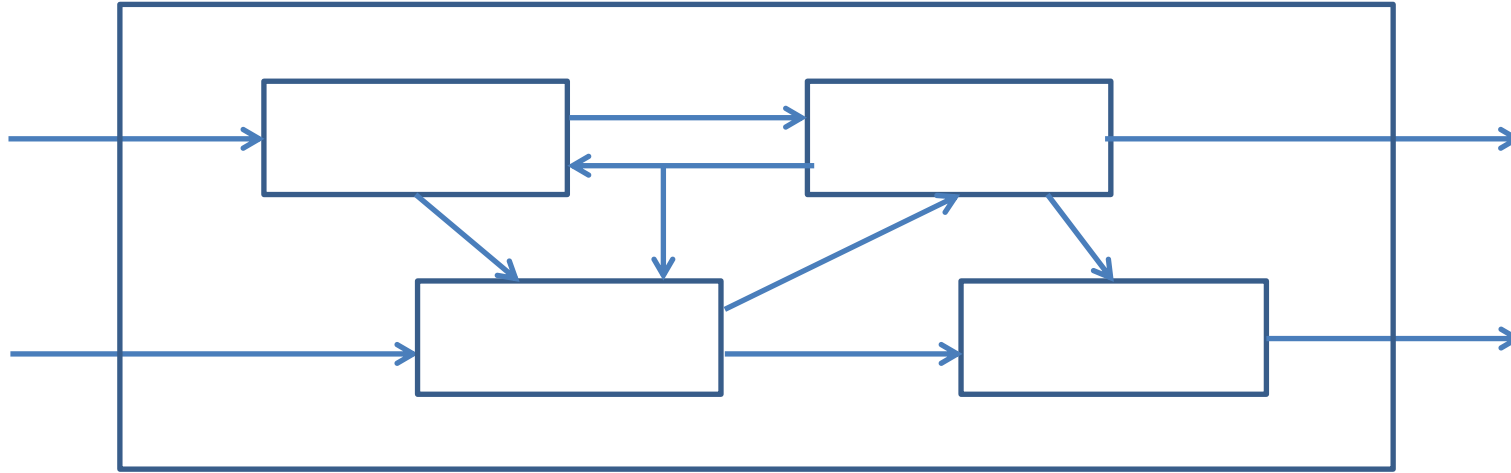


Each **mode switch** corresponds to a **task**

Examples:

- Input task: $(mode = stable \ \& \ in = x) \rightarrow mode := unstable$
- Output task: $(mode = stable) \rightarrow out := x$
- Internal task: $(mode = unstable) \rightarrow \{ x := \neg x ; mode := stable \}$
- ...

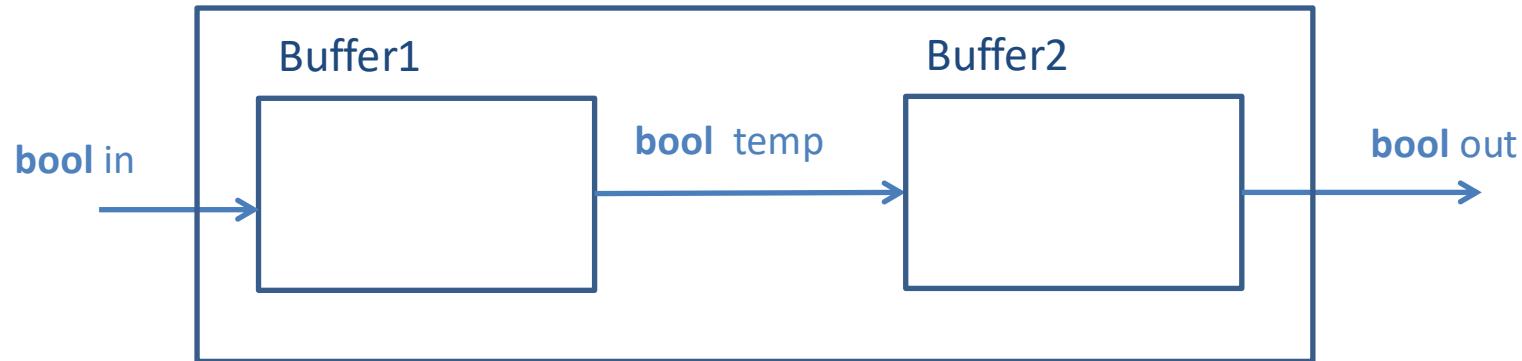
Block Diagrams



Visually the same as the synchronous case

However, their execution semantics is **different!**

DoubleBuffer



$(\text{Buffer}[\text{out} \mapsto \text{temp}] \mid \text{Buffer}[\text{in} \mapsto \text{temp}]) \setminus \text{temp}$

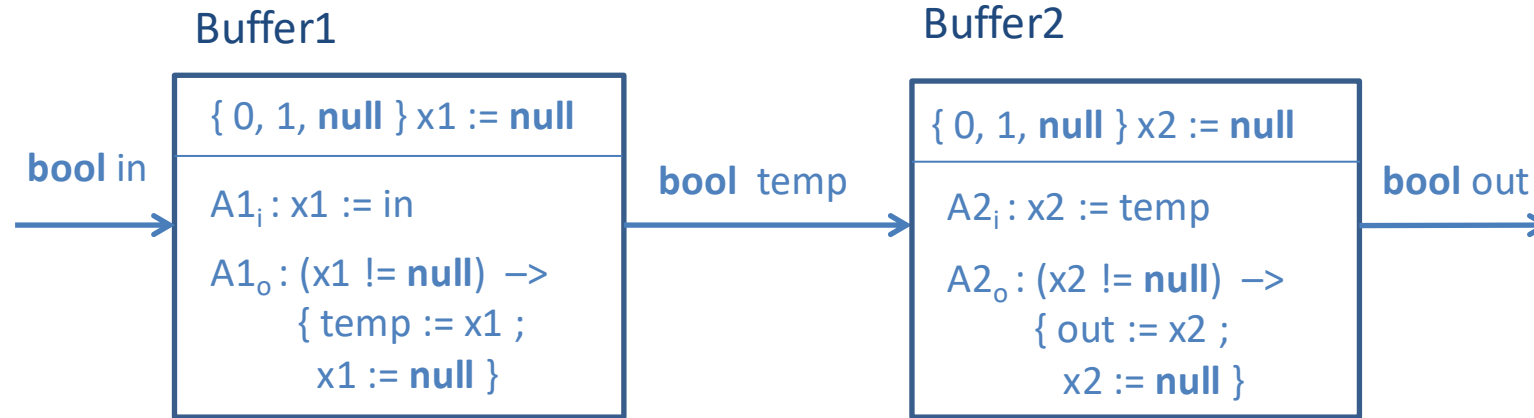
Instantiation: Create two instances of **Buffer**

- output of **Buffer1** = input of **Buffer2** = variable **temp**

Parallel composition: Asynchronous concurrent execution of **Buffer1** and **Buffer2**

Variable hiding: Encapsulation (**temp** becomes local)

Composing Buffer1 and Buffer2

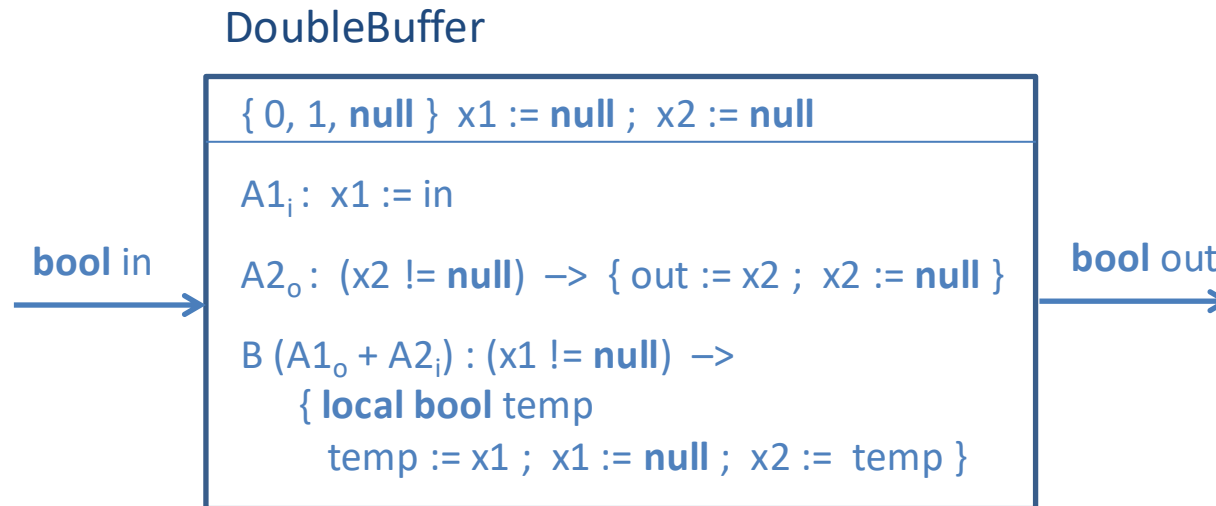
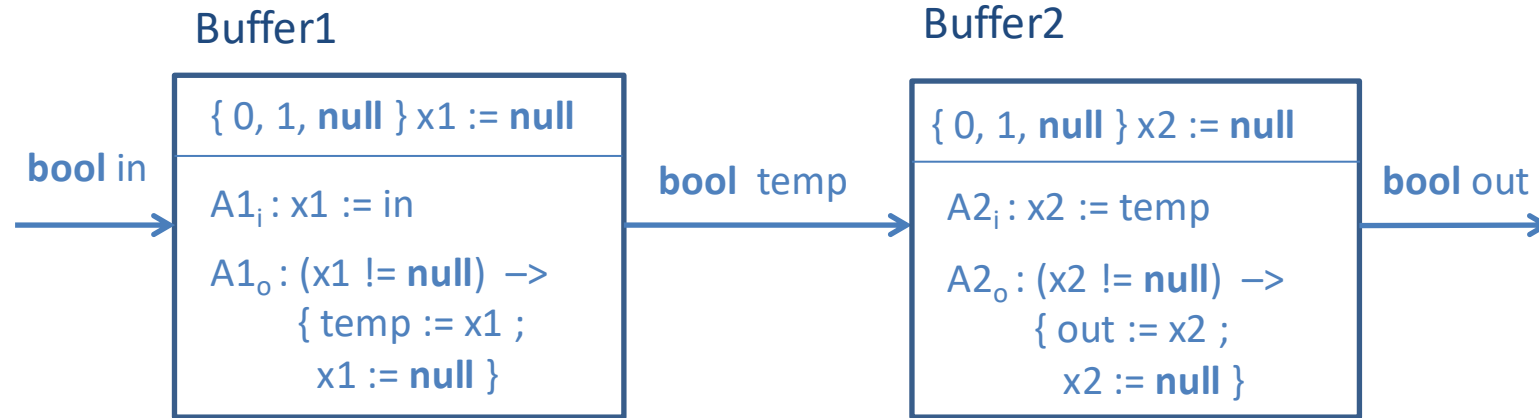


Inputs, outputs, states, and initialization for composition obtained as in synchronous case

What are the tasks of the composition?

- **Production** of output on **temp** by **Buffer1** **synchronized** with **consumption** of input on **temp** by **Buffer2**

Compiled DoubleBuffer



Asynchronous Composition (1)

Given asynchronous processes P_1 and P_2 , how to define $P_1 \mid P_2$?

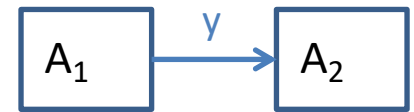
In each step of execution, **only one task** is executed

- Concepts such as await-dependencies, compatibility of interfaces are not relevant
- It is enough for P_1 and P_2 have **no output** channels **in common**

Example Case 1: (see textbook for complete definition)

If

- y is an **output** channel of P_1 and **input** channel of P_2
- A_1 is an **output** task of P_1 for y with code: $\text{Guard}_1 \rightarrow \text{Update}_1$
- A_2 is an **input** task of P_2 for y with code: $\text{Guard}_2 \rightarrow \text{Update}_2$



then

- $P_1 \mid P_2$ has an **output** task for y with code: $(\text{Guard}_1 \ \& \ \text{Guard}_2) \rightarrow \text{Update}_1 ; \text{Update}_2$

Asynchronous Composition (2)

Given asynchronous processes P_1 and P_2 , how to define $P_1 \mid P_2$?

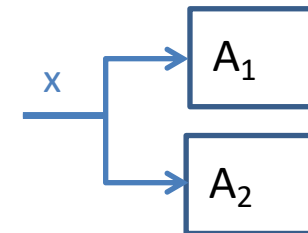
In each step of execution, **only one task** is executed

- Concepts such as await-dependencies, compatibility of interfaces are not relevant
- It is enough for P_1 and P_2 have **no output** channels **in common**

Example Case 2: (see textbook for complete definition)

If

- x is an **input** channel of P_1 and **input** channel of P_2
- A_1 is an **input** task of P_1 for x with code: $\text{Guard}_1 \rightarrow \text{Update}_1$
- A_2 is an **input** task of P_2 for x with code: $\text{Guard}_2 \rightarrow \text{Update}_2$



then

- $P_1 \mid P_2$ has an **input** task for x with code: $(\text{Guard}_1 \ \& \ \text{Guard}_2) \rightarrow \text{Update}_1 ; \text{Update}_2$

Asynchronous Composition (3)

Given asynchronous processes P_1 and P_2 , how to define $P_1 \mid P_2$?

In each step of execution, **only one task** is executed

- Concepts such as await-dependencies, compatibility of interfaces are not relevant
- It is enough for P_1 and P_2 have **no output** channels **in common**

Example Case 3: (see textbook for complete definition)

If

- x is an **input** channel of P_1 and **not** a channel of P_2
- A_1 is an **input** task of P_1 for x with code: $\text{Guard}_1 \rightarrow \text{Update}_1$



then

- $P_1 \mid P_2$ has an **input** task for x with code: $\text{Guard}_1 \rightarrow \text{Update}_1$

Asynchronous Composition (4)

Given asynchronous processes P_1 and P_2 , how to define $P_1 \mid P_2$?

In each step of execution, **only one task** is executed

- Concepts such as await-dependencies, compatibility of interfaces are not relevant
- It is enough for P_1 and P_2 have **no output** channels **in common**

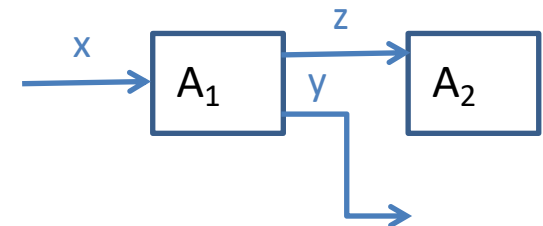
Example Case 4: (see textbook for complete definition)

If

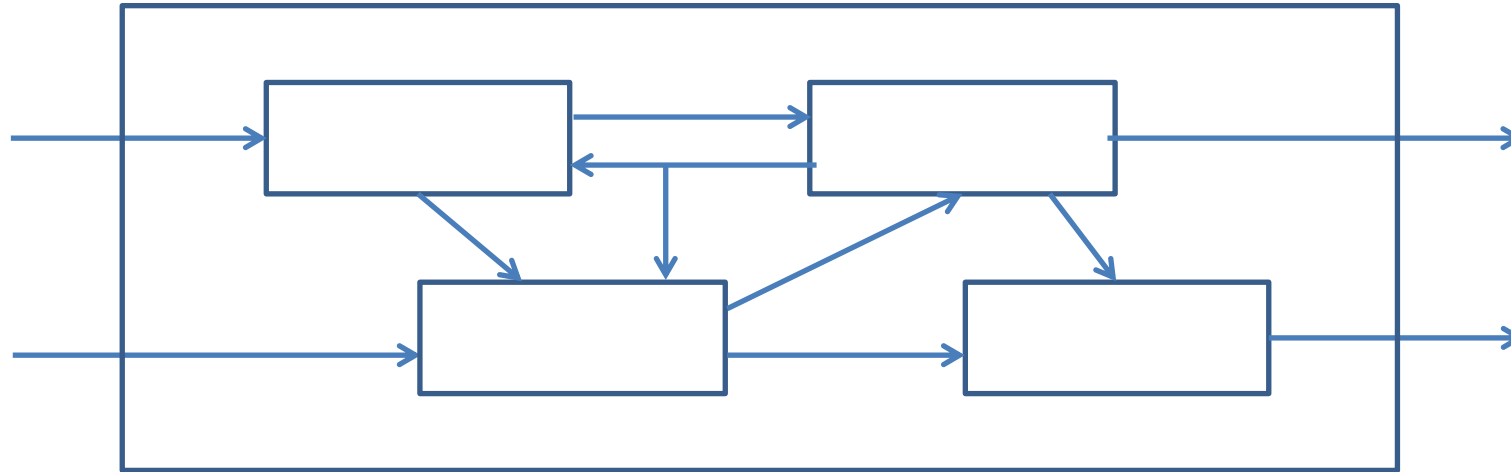
- y is an **output** channel of P_1 and **not** a channel of P_2
- A_1 is an **output** task of P_1 for y with code: $\text{Guard}_1 \rightarrow \text{Update}_1$

then

- $P_1 \mid P_2$ has an **output** task for y with code: $\text{Guard}_1 \rightarrow \text{Update}_1$



Execution Model: Another View



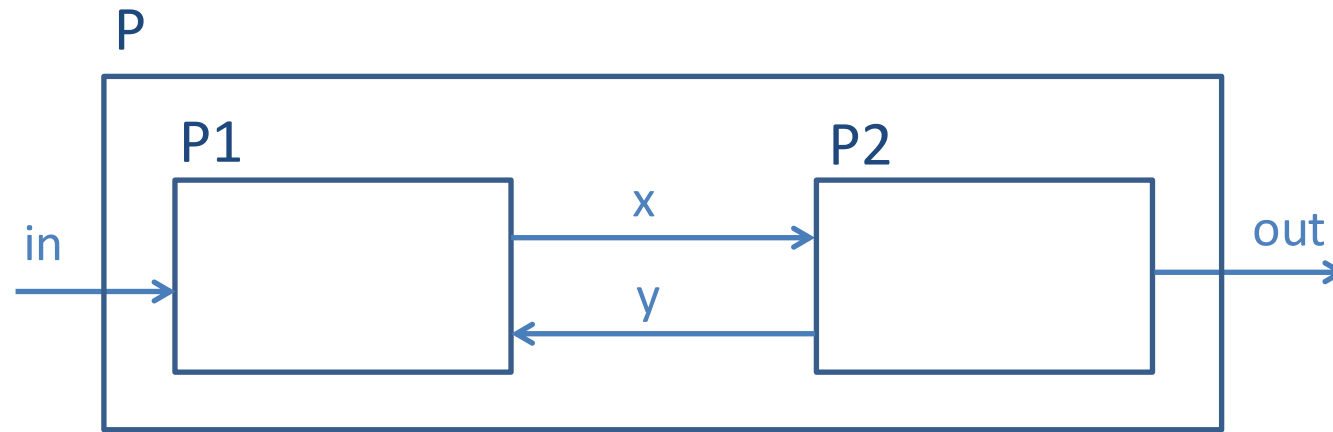
A single execution step:

- Execute an **internal task** of one of the processes, or
- **Process input** on an external channel x : execute an **input task** for x of every process to which x is an input, or
- Execute an **output task** for an output y of some process, followed by an **input task** for y for every process to which y is an input

If **multiple** choices are **enabled**, choose one **non-deterministically**

- No constraints on relative execution speeds

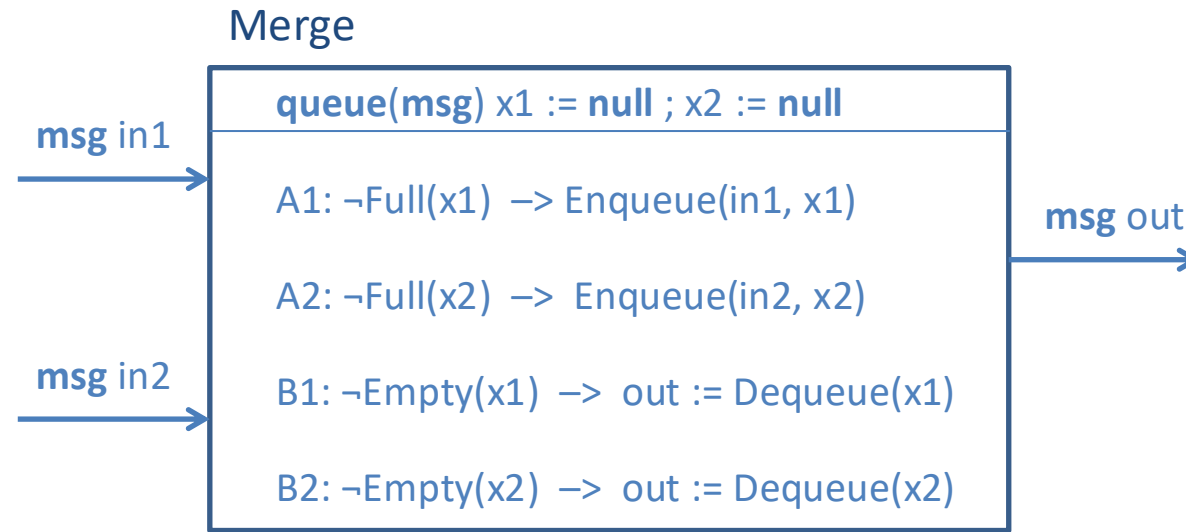
Asynchronous Execution



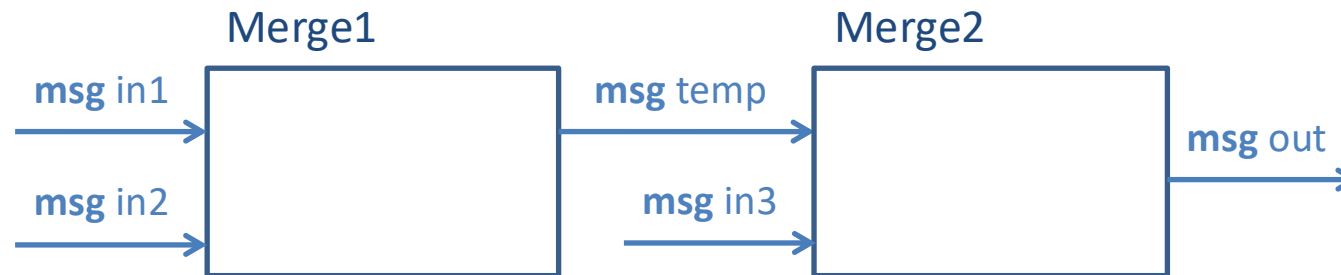
What can happen in a **single round** of this asynchronous model P ?

- $P1$ **synchronizes with the environment** to accept input on in
- $P2$ **synchronizes with the environment** to send output on out
- $P1$ **performs some internal computation** (one of its internal tasks)
- $P2$ **performs some internal computation** (one of its internal tasks)
- $P1$ **produces output** on x , followed by its immediate consumption by $P2$
- $P2$ **produces output** on y , followed by its immediate consumption by $P1$

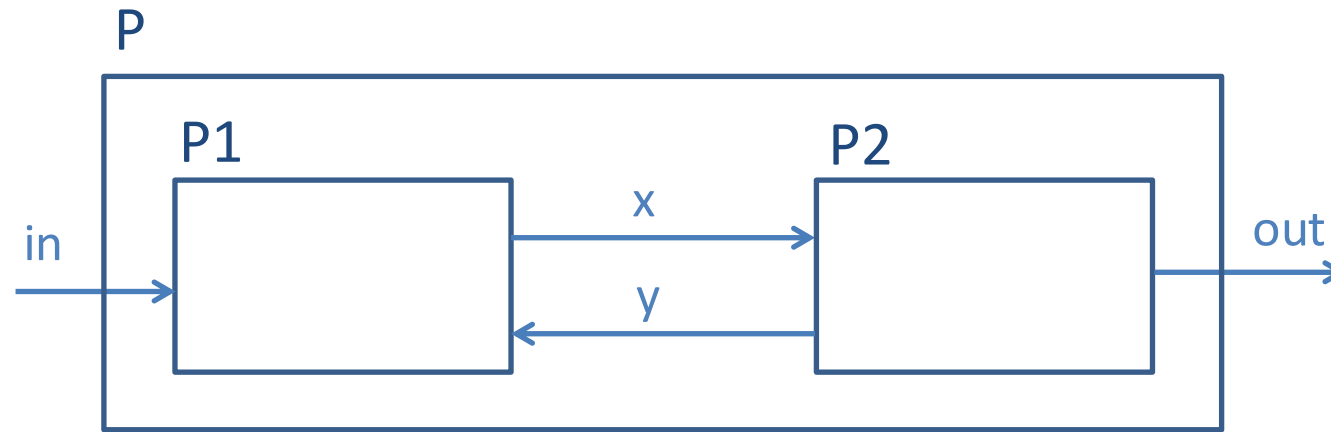
Asynchronous Merge



$\text{Merge}[\text{out} \mapsto \text{temp}] \mid \text{Merge}[\text{in1} \mapsto \text{temp}][\text{in2} \mapsto \text{in3}]$



Asynchronous Execution



Note: Inter-process communication is *blocking*:

if no task of $P2$ associated with x is enabled in a round, $P1$ cannot write to x in that round

A process P is *non-blocking* if for every input channel in and state s of P , some task of P associated with in is enabled in state s

In designs with *non-blocking* processes, a receiving process is often expected to send an *acknowledgement* back to the sender of a message m that it did receive m

Deadlocks

A commonly occurring error in **asynchronous** designs

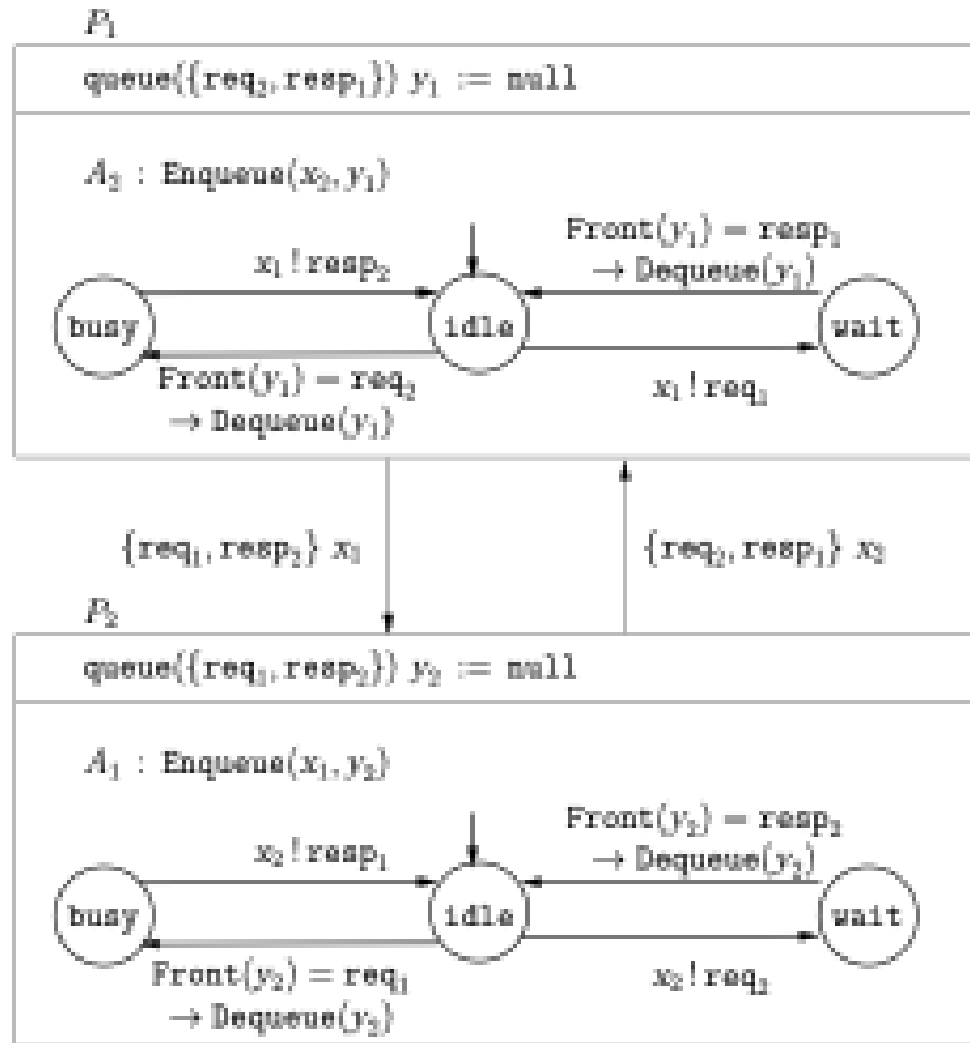
A **deadlock** occurs when **each process** is **waiting** for some other process to execute a task, but **no task** is **enabled**

In general, a state **s** of an asynchronous process **P** is a ***deadlock state*** if

1. **no task** is **enabled** in **s** and
2. **s** does **not** correspond to a **successful termination** of the system

The **absence** of **deadlocks** is a generic **safety requirement** that is expected to be an invariant of all asynchronous designs

Deadlock Example



Process P_i can generate a request that is serviced by process P_j ($i \neq j$), which provides a response to P_i 's request

Consider this transition of $P_1 \mid P_2$ for state $(P_1.\text{mode}, y_1, P_2.\text{mode}, y_2)$:

$(\text{idle}, \text{null}, \text{idle}, \text{null}) \xrightarrow{x_1! \text{req}_1} (\text{wait}, \text{null}, \text{idle}, [\text{req}_1])$

Two tasks are enabled:

1. mode switch from **idle** to **busy** in P_2
2. mode switch from **idle** to **wait** in P_2

In the second case, we get

$(\text{wait}, \text{null}, \text{idle}, [\text{req}_1]) \xrightarrow{x_2! \text{req}_2} (\text{wait}, [\text{req}_2], \text{wait}, [\text{req}_1])$

Now, no task is enabled: **deadlock!**

Credits

Notes based on Chapter 4 of

Principles of Cyber-Physical Systems

by Rajeev Alur

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